

4. Non-Functional Requirements

4.1 Performance

Load Testing: The system should handle 100 simultaneous users without performance degradation.

Response Times: Home page load within 3 seconds, real-time updates on parameter changes.

4.2 Usability

User Testing: Regular usability testing with target user groups to refine the interface, particularly regarding accessibility.

5. Validation Strategy

5.1 Validation Techniques

Unit Testing: For individual components.

Integration Testing: For interactions between components.

User Acceptance Testing (UAT): Involving end-users to validate against SRS.

5.2 Success Criteria

All critical requirements must be verified.

User feedback must indicate satisfactory usability and engagement.

6. Conclusion

This SDD outlines the architectural and design approach for the NASA Psyche Mission Simulator, providing a clear roadmap for development and implementation. By adhering to these specifications, the project aims to deliver a robust and engaging educational tool for various user groups.